YAKEEN NEET 2.0

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The **p-Block Elements**

Inorganic Chemistry | Lecture - 02

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Topics to be covered

1. Atomic and Physical Properties & NCERT lines)
2. Oxidation states and Chemical Reactivity (Group Wise)

NEET syllabus of p Block

UNIT 10: P- BLOCK ELEMENTS

Group -13 to Group 18 Elements

General Introduction: Electronic configuration and general trends in physical and chemical properties of elements across the periods and down the groups; unique behaviour of the first element in each group.

Atomic and physical properties (13th to 18th)

- Size
- I.E
- EN
- E.A (16th, 17th, 18th) (Imp)

- B.P
- M.P
- Density

Top to bottom orders

Size

14th to 18th
Top to bottom
Size ↑se
(No Exception)

group 13th

B
∧
Al
∨
Ga — $3d^{10}$ e⁻, σ↓ Zeff↑se, Size↓
∧
In (T.C)
∧
Tl

B < Ga ≤ Al < In < Tl

I.E.

15th to 18th
top to bottom ↓se

13th	14th
B	C
∨	∨
Al (T.C)	Si
∨	∨
Ga	Ge
∨	∨
In	Sn
∨	∨
Tl	Pb

↓
due to lanthanoid Contraction

Final order
13th B > Tl > Ga > Al > In
Beet — GAI

14th C > Si > Ge > Pb > Sn

M.P. & B.P.

"Experimental Values"
(Class 11th)

B.P
13th & 14th group — Top to bottom B.P ↓se
15th to 18th (Class 12th) — Top to bottom B.P ↑se
[* Sb > Bi * Te > Po]

M.P
(BATIn G)
13th B > Al > Tl > In > Ga *
(Same As I.E)
14th C > Si > Ge > Pb > Sn *
Nahi. Piyenge Sab Achhe Hai
15th N < P < Bi < Sb < As
Bidi
16th O < S < Se < Po < Te
17th F_2 < Cl_2 < Br_2 < I_2
18th He < Ne < Ar < Kr < Xe
] ↑se

E.N. Pauling scale

13th	14th	15th to 17th
B 2.0	C 2.5	Top to bottom ↓se
Al 1.5	Si 1.8	
Ga 1.6	Ge 1.8	
In 1.7	Sn 1.8	
Tl 1.8	Pb 1.9	

Aiims
13th EN Values largest gap b/w B & Al
2.0 — 1.5

Density

Top to bottom
↑se

* $C_{diamond}$ > Si

Table 11.2 Atomic and Physical Properties of Group 13 Elements

Property		Element				
		Boron B	Aluminium Al	Gallium Ga	Indium In	Thallium Tl
Atomic number		5	13	31	49	81
Atomic mass(g mol^{-1})		10.81	26.98	69.72	114.82	204.38
Electronic Configuration		$[He]2s^2 2p^1$	$[Ne]3s^2 3p^1$	$[Ar]3d^{10}4s^2 4p^1$	$[Kr]4d^{10}5s^2 5p^1$	$[Xe]4f^{14}5d^{10}6s^2 6p^1$
Atomic radius/pm[a]		(88)	143	135	167	170
Ionic radius M^{3+}/pm[b]		(27)	53.5	62.0	80.0	88.5
Ionic radius M^+/pm		-	-	120	140	150
Ionization enthalpy (kJ mol^{-1})	$\Delta_i H_1$	801	577	579	558	589
	$\Delta_i H_2$	2427	1816	1979	1820	1971
	$\Delta_i H_3$	3659	2744	2962	2704	2877
Electronegativity[c]		2.0	1.5	1.6	1.7	1.8
Density /g cm^{-3} at 298 K		2.35	2.70	5.90	7.31	11.85
Melting point / K		2453	933	303	430	576
Boiling point / K		3923	2740	2676	2353	1730
$E^{\ominus}$/ V for (M^{3+}/M)		-	–1.66	–0.56	–0.34	+1.26
$E^{\ominus}$ / V for (M^+/M)		-	+0.55	-0.79(acid) –1.39(alkali)	–0.18	–0.34

Ionic size M^{3+}
No exception
$B^{+3} < Al^{3+} < Ga^{3+} < In^{3+} < Tl^{3+}$

Some Important Points : NCERT Lines

Group : 13th

- Atomic radius of Ga is less than that of Al. ∵ T.C

- The observed discontinuity in the ionisation enthalpy values between Al and Ga, and between In and Tl are due to inability of d- and f-electrons.

  3d e⁻ σ↓ IE↑ T.C

  (4f e⁻ σ↓) L.C

  (Poor shielding)

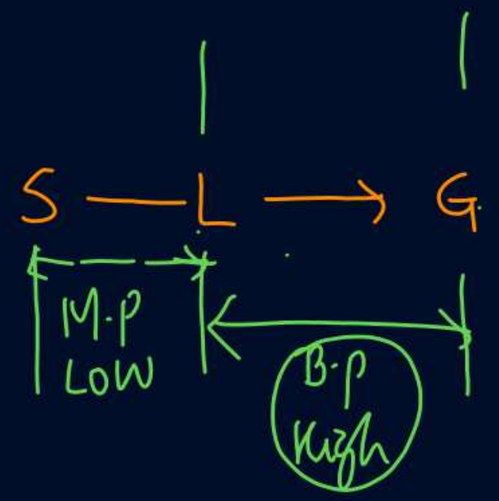
- Down the group, electronegativity first decreases from B to Al and then increases marginally.

  2.0 1.5

- Due to very strong crystalline lattice, boron has unusually high melting point. Rest of the members are soft metals with low melting point and high electrical conductivity.

Max B > Al > Tl > In > Ga Min

- It is worthwhile to note that gallium with unusually low melting point (303K), could exist in liquid state during summer. Its high boiling point (2676K) makes it a useful material for measuring high temperatures.



C < Si < Ge < Sn < Pb (size)
de⁻ & fe⁻

Group : 14th

- There is a considerable increase in covalent radius from C to Si, thereafter from Si to Pb a small increase in radius is observed.

- The first ionization enthalpy of group 14 members is higher than the corresponding members of group 13. $np^1 < np^2$ (13th, 14th) ↑se in Zeff ↓se in size Reason (L→R)

- Small decrease in $\Delta_i H$ from Si to Ge to Sn and slight increase in $\Delta_i H$ from Sn to Pb is the consequence of poor shielding effect of intervening d and f orbitals.

C > Si > Ge > Sn < Pb
small ↓se L.C

- All members of group14 are solids. Carbon and silicon are non-metals, germanium is a metalloid, whereas tin and lead are soft metals with low melting points. Melting points and boiling points of group 14 elements are much higher than those of corresponding elements of group 13.

Table 11.3 Atomic and Physical Properties of Group 14 Elements

Property		Element				
		Carbon C	Silicon Si	Germanium Ge	Tin Sn	Lead Pb
Atomic Number		6	14	32	50	82
Atomic mass (g mol^{-1})		12.01	28.09	72.60	118.71	207.2
Electronic configuration		$[He]2s^2 2p^2$	$[Ne]3s^2 3p^2$	$[Ar]3d^{10} 4s^2 4p^2$	$[Kr]4d^{10} 5s^2 5p^2$	$[Xe]4f^{14} 5d^{10} 6s^2 6p^2$
Covalent radius/pm[a]		77	118	122	140	146
Ionic radius M^{4+}/pm[b]		–	40	53	69	78
Ionic radius M^{2+}/pm[b]		–	–	73	118	119
Ionization enthalpy/ kJ mol^{-1}	$\Delta_i H_1$	1086	786	761	708	715
	$\Delta_i H_2$	2352	1577	1537	1411	1450
	$\Delta_i H_3$	4620	3228	3300	2942	3081
	$\Delta_i H_4$	6220	4354	4409	3929	4082
Electronegativity[c]		2.5	1.8	1.8	1.8	1.9
Density[d]/g cm^{-3}		3.51[e]	2.34	5.32	7.26[f]	11.34
Melting point/K		4373	1693	1218	505	600
Boiling point/K		–	3550	3123	2896	2024
Electrical resistivity/ ohm cm (293 K)		10^{14}–10^{16}	50	50	10^{-5}	2×10^{-5}

Group : 15th

Size → Top to bottom ↑se
IE → — " — ↓se

I.E L→R
Particular Period.

$p^2 < p^3 > p^4$

Half filled stability (I.E) more

- Dinitrogen is a diatomic gas while all others are solids. Metallic character increases down the group. Nitrogen and phosphorus are non-metals, arsenic and antimony metalloids and bismuth is a metal. This is due to decrease in ionisation enthalpy and increase in atomic size.

T→B
I.E↓
Met. ch↑

- The boiling points, in general, increase from top to bottom in the group but the melting point increases upto arsenic and then decreases upto bismuth. Except nitrogen, all the elements show allotropy.

∴ Low B.D.E of N–N Bond (l.p–l.p Repulsion) | Self linking very less

N < P < As > Sb > Bi

Final N < P < Bi < Sb < As

Table 7.1: Atomic and Physical Properties of Group 15 Elements

Property		N	P	As	Sb	Bi
Atomic number		7	15	33	51	83
Atomic mass/g mol^{-1}		14.01	30.97	74.92	121.75	208.98
Electronic configuration		$[He]2s^2 2p^3$	$[Ne]3s^2 3p^3$	$[Ar]3d^{10} 4s^2 4p^3$	$[Kr]4d^{10} 5s^2 5p^3$	$[Xe]4f^{14} 5d^{10} 6s^2 6p^3$
Ionisation enthalpy	I	1402	1012	947	834	703
($\Delta_i H$/(kJ mol^{-1}))	II	2856	1903	1798	1595	1610
	III	4577	2910	2736	2443	2466
Electronegativity		3.0	2.1	2.0	1.9	1.9
Covalent radius/pm[a]		70	110	121	141	148
Ionic radius/pm		171[b]	212[b]	222[b]	76[c]	103[c]
Melting point/K		63*	317[d]	1089[e]	904	544
Boiling point/K		77.2*	554[d]	888[f]	1860	1837
Density/[g cm^{-3}(298 K)]		0.879[g]	1.823	5.778[h]	6.697	9.808

E^{-3}

E^{3+}

Group : 16th

- The elements of this group have lower ionisation enthalpy values compared to those of Group15 in the corresponding periods. $np^3 > np^4$

Notes P.T

- Because of the compact nature of oxygen atom, it has less negative electron gain enthalpy than sulphur. However, from sulphur onwards the value again becomes less negative upto polonium. O < S (EA) S > Se > Te > Po > O](Final order)

- The metallic character increases from oxygen to polonium.

- Oxygen and sulphur are non-metals, selenium and tellurium metalloids, whereas polonium is a metal.

JEE Main PYQ

V.Imp

- The large difference between the melting and boiling points of oxygen and sulphur may be explained on the basis of their atomicity; oxygen exists as diatomic molecule (O_2) whereas sulphur exists as polyatomic molecule (S_8).

Table 7.6: Some Physical Properties of Group 16 Elements

Property	O	S	Se	Te	Po
Atomic number	8	16	34	52	84
Atomic mass/g mol^{-1}	16.00	32.06	78.96	127.60	210.00
Electronic configuration	$[He]2s^2 2p^4$	$[Ne]3s^2 3p^4$	$[Ar]3d^{10} 4s^2 4p^4$	$[Kr]4d^{10} 5s^2 5p^4$	$[Xe]4f^{14} 5d^{10} 6s^2 6p^4$
Covalent radius/(pm)[a]	66	104	117	137	146
Ionic radius, E^{2-}/pm	140	184	198	221	230[b]
Electron gain enthalpy, $/\Delta_{eg}H$ kJ mol^{-1}	–141	–200	–195	–190	–174
Ionisation enthalpy ($\Delta_i H_1$) /kJ mol^{-1}	1314	1000	941	869	813
Electronegativity	3.50	2.58	2.55	2.01	1.76
Density /g cm^{-3} (298 K)	1.32[c]	2.06[d]	4.19[e]	6.25	–
Melting point/K	55	393[f]	490	725	520
Boiling point/K	90	718	958	1260	1235
Oxidation states*	–2,–1,1,2	–2,2,4,6	–2,2,4,6	–2,2,4,6	2,4] later

Group : 17th

L→R Size ↓se Li>Be>B>C>N>O>F

- The halogens have the smallest atomic radii in their respective periods due to maximum effective nuclear charge.

- Atomic and ionic radii increase from fluorine to iodine due to increasing number of quantum shells. F<Cl<Br<I
  F⁻<Cl⁻<Br⁻<I⁻

- They have little tendency to lose electron. Thus they have very high ionisation enthalpy. Due to increase in atomic size, ionisation enthalpy decreases down the group. High IE I.E F>Cl>Br>I

- Halogens have maximum negative electron gain enthalpy in the corresponding periods. ∵ after addition of 1e⁻ they achieve Inert gas Configuration
  $np^5 + 1e^- \rightarrow np^6$

- Fluorine is the most electronegative element in the periodic table.
  E.N.O

P.T NOTE

Note :

The negative electron gain enthalpy of fluorine is less than that of chlorine. It is due to small size of fluorine atom. As a result, there are strong interelectronic repulsions in the relatively small 2p orbitals of fluorine and thus, the incoming electron does not experience much attraction.

Cl > F > Br > I (Most Repeated order)

$F_2 < Cl_2 < Br_2 < I_2$ (M.W ↑se, L.D.F ↑, Attraction ↑se b/w Molecules)

- Fluorine and chlorine are gases, bromine is a liquid and iodine is a solid.

- All halogens are coloured. This is due to absorption of radiations in visible region which results in the excitation of outer electrons to higher energy level.

HOMO–LUMO Transition

F_2 18e⁻

π^*2p_x, π^*2p_y (HOMO), σ^*2p_z (LUMO)

- F_2, has yellow, Cl_2, greenish yellow, Br_2, red and I_2, violet colour.

- Fluorine and chlorine react with water. Bromine and iodine are only sparingly soluble in water but are soluble in various organic solvents such as chloroform, carbon tetrachloride, carbon disulphide and hydrocarbons to give coloured solutions.

CCl_4 CS_2 $\mu = 0$ Non polar solvent

- Bond dissociation enthalpy order $Cl_2 > Br_2 > F_2 > I_2$.

l.p.- l.p Repulsion

Table 7.8: Atomic and Physical Properties of Halogens

Property	F	Cl	Br	I	At[a]
Atomic number	9	17	35	53	85
Atomic mass/g mol^{-1}	19.00	35.45	79.90	126.90	210
Electronic configuration	$[He]2s^2 2p^5$	$[Ne]3s^2 3p^5$	$[Ar]3d^{10} 4s^2 4p^5$	$[Kr]4d^{10} 5s^2 5p^5$	$[Xe]4f^{14} 5d^{10} 6s^2 6p^5$
Covalent radius/pm	64	99	114	133	–
Ionic radius X^-/pm	133	184	196	220	–
Ionisation enthalpy/kJ mol^{-1}	1680	1256	1142	1008	–
Electron gain enthalpy/kJ mol^{-1}	–333	–349	–325	–296	–
Electronegativity[b]	4	3.2	3.0	2.7	2.2
$\Delta_{Hyd}H(X^-)$/kJ mol^{-1}	515	381	347	305	–
	F_2	Cl_2	Br_2	I_2	–
Melting point/K	54.4	172.0	265.8	386.6	–
Boiling point/K	84.9	239.0	332.5	458.2	–
Density/g cm^{-3}	1.5 (85)[c]	1.66 (203)[c]	3.19(273)[c]	4.94(293)[d]	–
Distance X – X/pm	143	199	228	266	–
Bond dissociation enthalpy /(kJ mol^{-1})	158.8	242.6	192.8	151.1	–
$E^{\ominus}/V$[e]	2.87	1.36	1.09	0.54	–

Group : 18th

(np6) max in period

- Due to stable electronic configuration these gases exhibit very high ionisation enthalpy. However, it decreases down the group with increase in atomic size.
- Large positive values of electron gain enthalpy. PYQ
- All the noble gases are monoatomic.
- Colourless, odourless and tasteless and sparingly soluble in water.
- Very low melting and boiling points because the only type of interatomic interaction in these elements is weak dispersion forces. PYQ
- Helium has the lowest boiling point (4.2 K) of any known substance. Helium has an unusual property of diffusing through most commonly used laboratory materials such as rubber, glass or plastics.

Table 7.12: Atomic and Physical Properties of Group 18 Elements

Propery	He	Ne	Ar	Kr	Xe	Rn*
Atomic number	2	10	18	36	54	86
Atomic mass/ g mol^{-1}	4.00	20.18	39.95	83.80	131.30	222.00
Electronic configuration	$1s^2$	$[He]2s^22p^6$	$[Ne]\ 3s^23p^6$	$[Ar]3d^{10}4s^24p^6$	$[Kr]4d^{10}5s^25p^6$	$[Xe]4f^{14}5d^{10}6s^26p^6$
Atomic radius/pm	120	160	190	200	220	–
Ionisation enthalpy /$kJmol^{-1}$	2372	2080	1520	1351	1170	1037
Electron gain enthalpy /$kJmol^{-1}$	48	116	96	96	77	68
Density (at STP)/gcm^{-3}	1.8×10^{-4}	9.0×10^{-4}	1.8×10^{-3}	3.7×10^{-3}	5.9×10^{-3}	9.7×10^{-3}
Melting point/K	–	24.6	83.8	115.9	161.3	202
Boiling point/K	4.2	27.1	87.2	119.7	165.0	211
Atmospheric content (% by volume)	5.24×10^{-4}	1.82×10^{-3}	0.934	1.14×10^{-4}	8.7×10^{-6}	–

Oxidation state & trends in Chemical reactivity : Group 13th

PYQ IEE

- Boron does not exist as free B^{3+} ions.

Can not Exist as B^{3+} ions ← High Energy Very Small size

Reason : due to High Sum of three I.E.

$$B \xrightarrow{IE_1} B^{+} \xrightarrow{IE_2} B^{2+} \xrightarrow{IE_3} B^{3+}$$

Always Form Covalent Bond

- Stability of +1 and +3 oxidation states :

Inert pair effect ✓

done

But Al^{3+} Can Exist ∵ Low IE.

Al^{3+} { $3F^-$

Ionic

- In thallium +1 oxidation state is predominant whereas the +3 oxidation state is highly oxidising in character. done

$Tl^{+3} \rightarrow Tl^{+1}$
(O.A)

- The compounds in +1 oxidation state, as expected from energy considerations, are more ionic than those in +3 oxidation state.

Fajan's Rule.
Low O.S of cation more Ionic
+ve charge on cation ↑se
↓
Cov. ch ↑se

Problem 11.1

Standard electrode potential values, $E^{\ominus}$ for Al^{3+}/Al is -1.66 V and that of Tl^{3+}/Tl is $+1.26$ V. Predict about the formation of M^{3+} ion in solution and compare the electropositive character of the two metals.

Solution

Standard electrode potential values for two half cell reactions suggest that aluminium has high tendency to make $Al^{3+}(aq)$ ions, whereas Tl^{3+} is not only unstable in solution but is a powerful oxidising agent also. Thus Tl^{+} is more stable in solution than Tl^{3+}. Aluminium being able to form +3 ions easily, is more electropositive than thallium.

$Tl^{+1} > Tl^{+3}$ (stability)

[E ⇒ Elements]

(i) Reactivity towards air ($O_2 + N_2$)

$$E + O_2 \longrightarrow \overset{+3}{E_2}O_3$$

oxides

$$E + N_2 \longrightarrow \overset{+3}{E}N$$

Nitride

BN, AlN ... etc

Crystalline Boron
~~Reactive~~

Amorphous Boron.
Reactive ✓

O^{2-} of B ⇒ Acidic

O^{2-} of Al & Ga ⇒ Ampho.

O^{2-} of In & Tl ⇒ Basic

THANK YOU